UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 5 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages

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QUESTION 1

a) Solve the compound inequality -6 < -3x - 5 < 6. Express the answer in interval notation.

(4 marks)



Figure: solution on a number line.

b) Solve the inequality |-4x - 6| > 8. Express the answer in interval notation and on a number line.

(5 marks)

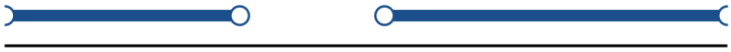


Figure: solution on a number line.

c) Sketch y = 3|x - 3| - 1. State the vertex, the y-intercept and any x-intercepts.

(5 marks)

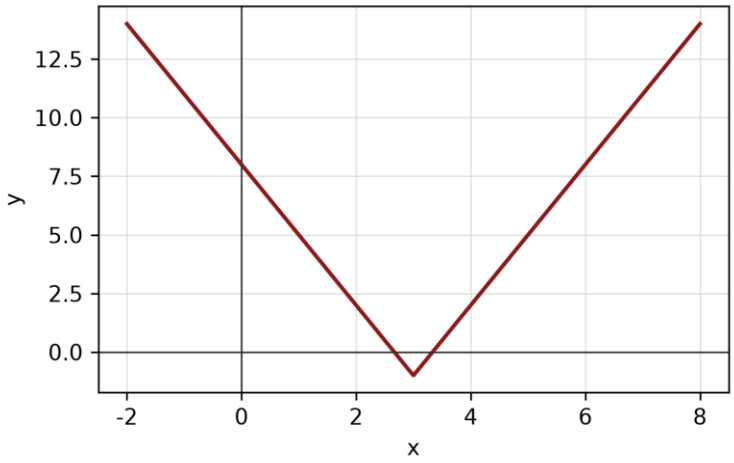


Figure: absolute-value graph.

QUESTION 2

a) Given f(x) = (6 - x)/(3x - 1), x ≠ 1/3, find $f^{-1}(x)$ and state its domain.

(5 marks)

b) Given f(x) = 5 - 3x and g(x) = 1/(4x + 4), find (g ∘ f)(x) and state its domain.

(5 marks)

c) Find the equation of the line with x-intercept -1 and y-intercept -8.

(4 marks)

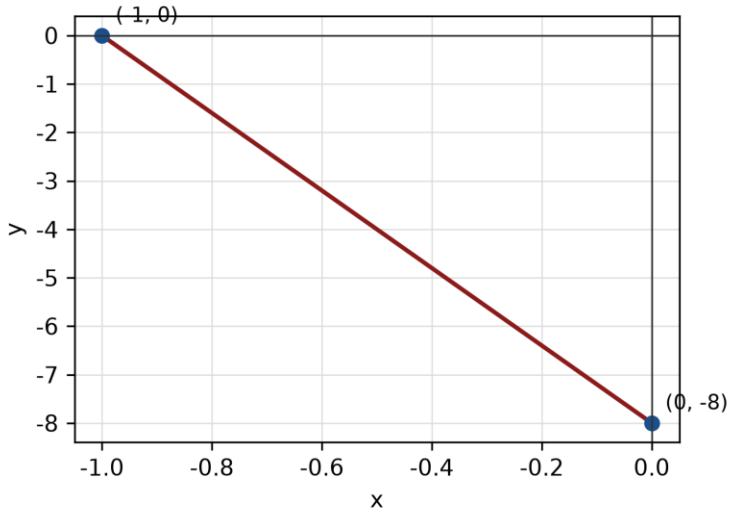


Figure: line and intercepts.

QUESTION 3

a) Given $z_1 = -4 - 5i$ and $z_2 = -6 - 4i$, find $z_1 + z_2$ in the form a + bi.

(4 marks)

b) Evaluate i^11 and simplify completely.

(3 marks)

c) Use the remainder theorem to find the remainder when f(x) = 2x^3 + 5x² + x + 3 is divided by x + 4.

(3 marks)

QUESTION 4

a) Solve the simultaneous equations
$y = x^2 + 6$
$y = x + 8$.

(5 marks)

b) Solve the system of equations simultaneously:
-4x + 3y - z = 9
2x - y - 2z = -1
-x - y - 4z = 15

(5 marks)

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c) Write the equation of the circle with centre (-3, -3) and radius 5. Hence write its general equation and sketch the circle.

(5 marks)

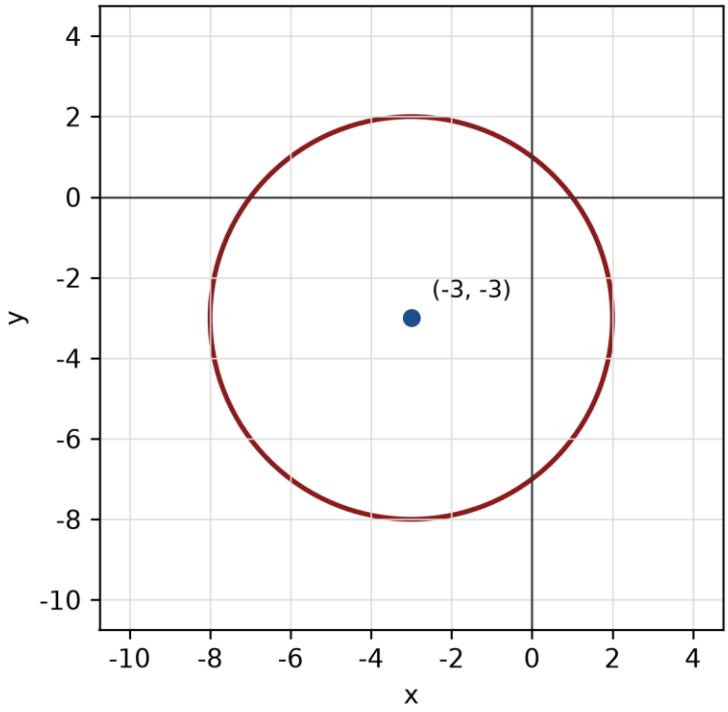


Figure: circle.

QUESTION 5

a) Prove the identity tan x + cos x/(1 + sin x) = sec x.

(5 marks)

b) A circle has radius 6 cm. Find the length of the arc subtended by a central angle of 120°.

(4 marks)

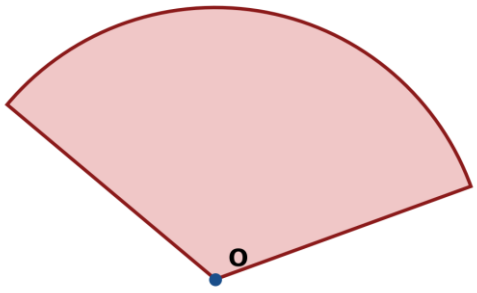


Figure: arc and sector.

c) In triangle ABC, b = 7 cm, c = 5 cm and A = 120°. Use the cosine rule to find a.

(4 marks)

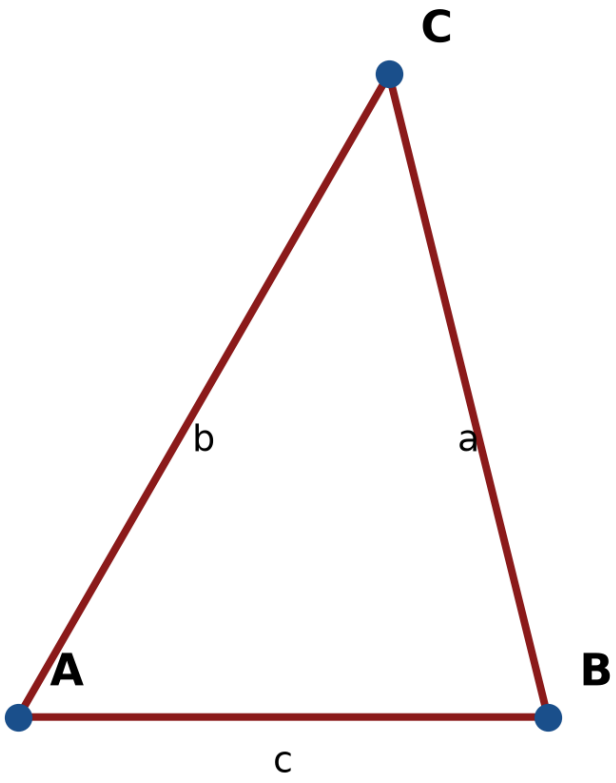


Figure: triangle ABC.

END OF QUESTION PAPER

APPENDIX 1

Formula sheet (as supplied with the paper).

$\sin^2 A + \cos^2 A = 1$
$\sin(A + B) = \sin A \cos B + \cos A \sin B$
$\cos(A + B) = \cos A \cos B - \sin A \sin B$
$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$
$\cos 2A = \cos^2 A - \sin^2 A$
$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$
$\sin 2A = 2 \sin A \cos A$
$\tan^2 A + 1 = \sec^2 A$
$\tan 2A = 2 \tan A / (1 - \tan^2 A)$
Arc length, $s = r\theta$
Area of sector = $\frac{1}{2}r^2\theta$
$a / \sin A = b / \sin B = c / \sin C$
$a^2 = b^2 + c^2 - 2bc \cos A$
Area of triangle = $\frac{1}{2}ab \sin C$

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